IB MATHEMATICS SL • UNIT 1.01

# C Topic 1: Number & Algebra (Sequences, Logs & Binomial)

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

**Total Questions:** 25 • Verified Step-by-Step Solutions

## QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

| | | | | |
|---|---|---|---|---|
| Q1 A | Q2 A | Q3 A | Q4 B | Q5 A |
| Q6 A | Q7 C | Q8 B | Q9 B | Q10 B |
| Q11 C | Q12 C | Q13 D | Q14 C | Q15 B |
| Q16 B | Q17 D | Q18 B | Q19 D | Q20 B |
| Q21 C | Q22 B | Q23 D | Q24 B | Q25 D |

## Topic 1: Number & Algebra (Sequences, Logs & Binomial) Practice Exercises

**1** MCQ

Consider the function $f(x) = 2x + 5$. What is the value of $f(3)$?

- **A** 11 ✓
- **B** 8
- **C** 10
- **D** 16

**WORKED SOLUTION & VERIFICATION**

To find $f(3)$, substitute $x = 3$ into the function $f(x) = 2x + 5$. $f(3) = 2(3) + 5 = 6 + 5 = 11$. **[M1] [A1]**

**2** MCQ

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown below. Find the value of the y-intercept.

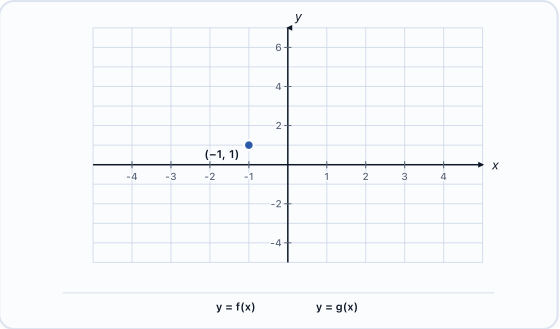
- **A** 5 ✓
- **B** -5
- **C** 0
- **D** 1

**WORKED SOLUTION & VERIFICATION**

The y-intercept of a function occurs where the graph crosses the y-axis, which is when $x = 0$. For the function $f(x) = x^2 - 6x + 5$, when $x = 0$, we have $f(0) = (0)^2 - 6(0) + 5 = 0 - 0 + 5 = 5$. Therefore, the y-intercept is 5. **[M1] [A1]**

**3** MCQ

Consider the function $f(x) = x^2 - 4x + 3$ and the line $g(x) = x - 1$. The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.




- **A** (1, 0) ✓
- **B** (3, 2)
- **C** (2, 1)
- **D** (4, 3)

**WORKED SOLUTION & VERIFICATION**

To find the intersection points, we set the two functions equal to each other: $f(x) = g(x)$ $x^2 - 4x + 3 = x - 1$ $x^2 - 5x + 4 = 0$ Factoring the quadratic equation: $(x - 1)(x - 4) = 0$ This gives us two possible x-values for the intersection points: $x = 1$ or $x = 4$. \nUsing $g(x) = x - 1$ to find the corresponding y-values: \nFor $x = 1$: $y = 1 - 1 = 0$. The intersection point is $(1, 0)$. \nFor $x = 4$: $y = 4 - 1 = 3$. The intersection point is $(4, 3)$. \nLooking at the graph, one of the intersection points is $(1, 0)$. **[M1] [A1]**

**4** MCQ

$e^0 =$

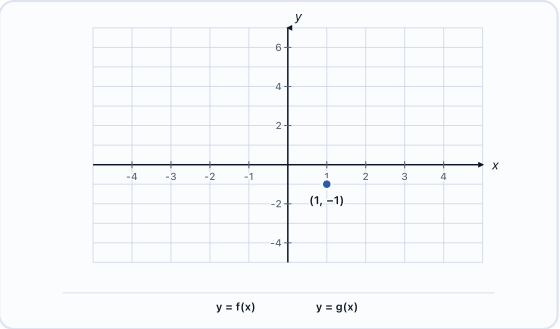
- **A** 0
- **B** 1 ✓
- **C** e
- **D** ∞

**WORKED SOLUTION & VERIFICATION**

Any nonzero number to power 0 is 1. **[M1] [A1]**

**5** MCQ

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. It passes through the points $(0, -3)$, $(1, 0)$, and $(-1, -4)$. Find the value of $c$.




- **A** -3 ✓
- **B** 3
- **C** -1
- **D** 1

**WORKED SOLUTION & VERIFICATION**

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The graph of the function passes through the point $(0, -3)$. Substituting these coordinates into the function, we get: $f(0) = a(0)^2 + b(0) + c$. Since $f(0) = -3$, we have $-3 = 0 + 0 + c$, which means $c = -3$. **[M1] [A1]**

**6** MCQ

Consider the function $f(x) = 2x - 3$. Determine the value of $f(3)$.

- **A** 3 ✓
- **B** 6
- **C** 9
- **D** 0

**WORKED SOLUTION & VERIFICATION**

To find the value of $f(3)$, substitute $x = 3$ into the function $f(x) = 2x - 3$.

$f(3) = 2(3) - 3$

$f(3) = 6 - 3$

$f(3) = 3$ **[M1] [A1]**

**7** MCQ

Solve for $x$: $3^x = 9$.

- **A** 3
- **B** 4
- **C** 2 ✓
- **D** 1

**WORKED SOLUTION & VERIFICATION**

Working through this algebra step: $3^x = 9 = (3)^2 \Rightarrow x = 2$. **[M1] [A1]**

**8** MCQ

An arithmetic sequence has first term $u_1 = 7$ and common difference $d = 2$. Find $u_{15}$.

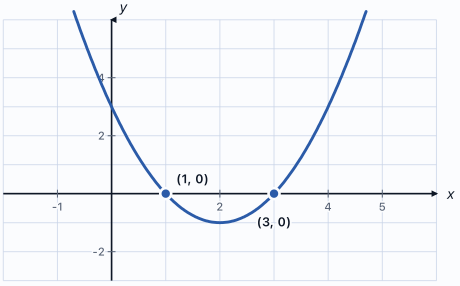
| | |
|---|---|
| A 37 | **B 35** ✓ |
| C 33 | D -21 |

**WORKED SOLUTION & VERIFICATION**

$u_n = u_1 + (n-1)d \Rightarrow u_{15} = (7) + (15-1)(2) = 35$. **[M1] [A1]**

**9** MCQ

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. What are the coordinates of the x-intercepts of the function?




| | |
|---|---|
| A (0, 1) and (0, 3) | **B (1, 0) and (3, 0)** ✓ |
| C (-1, 0) and (-3, 0) | D (0, -1) and (0, -3) |

**WORKED SOLUTION & VERIFICATION**

The x-intercepts of a function are the points where the graph crosses the x-axis, meaning the y-coordinate is 0. Thus, we need to solve $f(x) = 0$.

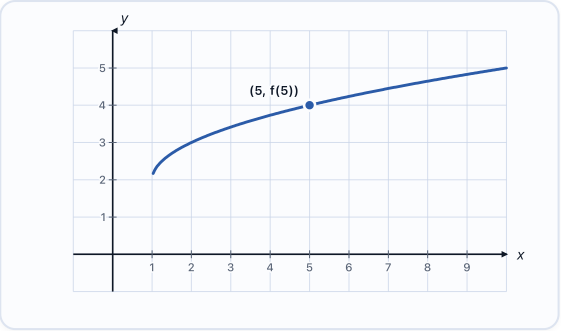
$x^2 - 4x + 3 = 0$

Factoring the quadratic equation, we get $(x-1)(x-3) = 0$

Therefore, the x-intercepts occur at $x = 1$ and $x = 3$. The coordinates of the x-intercepts are (1, 0) and (3, 0). **[M1] [A1]**

**10** MCQ

The graph of the function $f(x) = \sqrt{x-1} + 2$ is shown below. What is the value of $f(5)$?




A 2

B 4 ✓

C 3

D 5

**WORKED SOLUTION & VERIFICATION**

To find the value of $f(5)$, substitute $x = 5$ into the function $f(x) = \sqrt{x-1} + 2$.

$f(5) = \sqrt{5-1} + 2$

$f(5) = \sqrt{4} + 2$

$f(5) = 2 + 2$

$f(5) = 4$

Therefore, the value of $f(5)$ is 4. **[M1] [A1]**

**11** MCQ

An arithmetic sequence has first term $u_1 = 4$ and common difference $d = -1$. Find $u_{15}$.

A -11

B 18

C -10 ✓

D -9

**WORKED SOLUTION & VERIFICATION**

$u_n = u_1 + (n-1)d \Rightarrow u_{15} = (4) + (15-1)(-1) = -10$. **[M1] [A1]**

**12** MCQ

Evaluate $\log_{10}(10000)$.

A 2

B 3

C 4 ✓

D 1

**WORKED SOLUTION & VERIFICATION**

Working through this algebra step: $\log_{10}(10000) = x \Rightarrow 10^x = 10000 \Rightarrow x = 4$, since $10^4 = 10000$. **[M1] [A1]**

**13** MCQ

A geometric sequence has first term $u_1 = -4$ and common ratio $r = -3$. Find $u_7$.

| | |
|---|---|
| A 972 | B -2912 |
| C 8748 | D **-2916** ✓ |

**WORKED SOLUTION & VERIFICATION**

$u_n = u_1 r^{n-1} \Rightarrow u_7 = (-4)(-3)^{7-1} = -2916$. **[M1] [A1]**

**14** MCQ

Given $\log_5(6) = p$, express $\log_5(6^3)$ in terms of $p$.

| | |
|---|---|
| A $\frac{p}{3}$ | B $p^3$ |
| C $3p$ ✓ | D $p+3$ |

**WORKED SOLUTION & VERIFICATION**

Working through this algebra step: By the log law $\log(x^k) = k \log x$, the expression equals $3p$. **[M1] [A1]**

**15** MCQ

An arithmetic sequence has first term $u_1 = 4$ and common difference $d = -4$. Find $u_{19}$.

| | |
|---|---|
| A 76 | B **-68** ✓ |
| C -72 | D -64 |

**WORKED SOLUTION & VERIFICATION**

$u_n = u_1 + (n-1)d \Rightarrow u_{19} = (4) + (19-1)(-4) = -68$. **[M1] [A1]**

**16** MCQ

A geometric sequence has first term $u_1 = 5$ and common ratio $r = -3$. Find $u_6$.

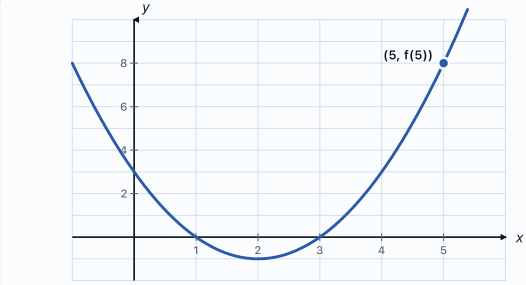
| | |
|---|---|
| A 3645 | B **-1215** ✓ |
| C 405 | D -1216 |

**WORKED SOLUTION & VERIFICATION**

$u_n = u_1 r^{n-1} \Rightarrow u_6 = (5)(-3)^{6-1} = -1215$. **[M1] [A1]**

**17** MCQ

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. Find the value of $f(5)$.




A 12

B 18

C 20

D **8** ✓

**WORKED SOLUTION & VERIFICATION**

To find $f(5)$, we substitute $x = 5$ into the function $f(x) = x^2 - 4x + 3$.

$f(5) = (5)^2 - 4(5) + 3$
$f(5) = 25 - 20 + 3$
$f(5) = 5 + 3$
$f(5) = 8$. **[M1] [A1]**

**18** MCQ

Determine the numerical value of $\log_5(125)$.

A 1

B **3** ✓

C 4

D 2

**WORKED SOLUTION & VERIFICATION**

Working through this algebra step: $\log_5(125) = x \Rightarrow 5^x = 125 \Rightarrow x = 3$, since $5^3 = 125$. **[M1] [A1]**

**19** MCQ

An arithmetic sequence has first term $u_1 = 3$ and common difference $d = 6$. Find $u_{19}$.

A 117

B 105

C -105

D **111** ✓

**WORKED SOLUTION & VERIFICATION**

$u_n = u_1 + (n-1)d \Rightarrow u_{19} = (3) + (19-1)(6) = 111$. **[M1] [A1]**

**20** MCQ

A geometric sequence has first term $u_1 = -2$ and common ratio $r = \frac{1}{3}$. Find $u_6$.

A 1

B $-\frac{2}{243}$ ✓

C $-\frac{2}{729}$

D $-\frac{2}{81}$

**WORKED SOLUTION & VERIFICATION**

$u_n = u_1 r^{n-1} \Rightarrow u_6 = (-2)(\frac{1}{3})^{6-1} = -\frac{2}{243}$. **[M1] [A1]**

**21** MCQ

A geometric sequence has first term $u_1 = -1$ and common ratio $r = -3$. Find $u_5$.

A 27

B -91

C **-81** ✓

D 243

**WORKED SOLUTION & VERIFICATION**

$u_n = u_1 r^{n-1} \Rightarrow u_5 = (-1)(-3)^{5-1} = -81$. **[M1] [A1]**

**22** MCQ

Find the root(s) satisfying for $x$: $5^x = 15625$.

A 3

B **6** ✓

C 2

D 4

**WORKED SOLUTION & VERIFICATION**

Working through this algebra step: $5^x = 15625 = (5)^6 \Rightarrow x = 6$. **[M1] [A1]**

**23** MCQ

Calculate the solution set for for $x$: $3^x = 81$.

A 3

B 1

C 2

D **4** ✓

**WORKED SOLUTION & VERIFICATION**

Working through this algebra step: $3^x = 81 = (3)^4 \Rightarrow x = 4$. **[M1] [A1]**

**24** MCQ

A geometric sequence has first term $u_1 = -1$ and common ratio $r = 3$. Find $u_6$.

| | |
|---|---|
| A -729 | B **-243** ✓ |
| C -253 | D -81 |

**WORKED SOLUTION & VERIFICATION**

$u_n = u_1 r^{n-1} \Rightarrow u_6 = (-1)(3)^{6-1} = -243$. **[M1] [A1]**

**25** MCQ

A geometric sequence has first term $u_1 = -2$ and common ratio $r = 2$. Find $u_5$.

| | |
|---|---|
| A -16 | B -31 |
| C -64 | D **-32** ✓ |

**WORKED SOLUTION & VERIFICATION**

$u_n = u_1 r^{n-1} \Rightarrow u_5 = (-2)(2)^{5-1} = -32$. **[M1] [A1]**